

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-a8911ee6e108230c8.jsonl`
- SHA-256: `f4a5a293263ac31d313a788121f445ecb900744f1b4271c9aa6172dac93554fa`
- Source modified: `2026-06-19T15:53:13+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T15:52:06.165Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.

HARD RULES:

- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.

Working dir is the repo root; use Read/Grep/Glob freely.

ADVERSARIALLY VERIFY these proposed staleness fixes for ONE file:

docs/MULTI_SIGNAL_FUSION_PLAN.md.

For EACH proposed fix: (1) open docs/MULTI_SIGNAL_FUSION_PLAN.md and confirm stale_text appears VERBATIM (tighten/expand it to an exact, UNIQUE match if needed ? this text will be applied by exact string match, so it must be perfect); (2) re-open the cited code yourself and confirm the claim is genuinely contradicted by the CURRENT code; (3) confirm corrected_text is accurate, preserves voice, and is minimal; (4) DROP it if it is subjective, an owner-gated decision, already-correct, or you cannot verify the evidence. Output ONLY the survivors (verdict confirm or revise), with stale_text guaranteed to be an exact unique substring of the file.

PROPOSED FIXES:

```
[
  {
    "file": "docs/MULTI_SIGNAL_FUSION_PLAN.md",
    "stale_text": "machine, future**).\`" The person cue in this plan **is that named-but-unbuilt Gate B.**",
    "corrected_text": "machine, future**).\`" The person cue in this plan **is that Gate B ? now built (default-OFF + unmeasured on golden) as the `FusionSegmenter` person path, see §10 P2.**",
    "why_stale": "Says Gate B is \"named-but-unbuilt\", but the person cue / Gate B is implemented in backend/pipeline/segmenters/fusion_hybrid.py (_select_for_handoff min_players gate at line 166; _person_features at lines 111-139) and backend/pipeline/detectors/fusion_features.py. The same doc's §10 P2 (line 271) already marks the FusionSegmenter DONE (2026-06-14), so §1 contradicts §10. It is default-OFF + unmeasured on golden, not unbuilt.",
    "severity": "medium"
  },
  {
    "file": "docs/MULTI_SIGNAL_FUSION_PLAN.md",
```

```

    "stale_text": "          ? (?? in-court players ? recent track) (Gate B ? the person
cue, NEW)",
    "corrected_text": "          ? (?? in-court players ? recent track) (Gate B ? the
person cue, BUILT default-OFF, unmeasured on golden)",
    "why_stale": "Tags Gate B \"NEW\" (not built). The occupancy gate is built:
FusionConfig.min_players (backend/config/models.py:108) is applied in
fusion_hybrid.py::_select_for_handoff (line 166), and players_in_court is computed in
fusion_features.py (PERSON_FEATURE_NAMES, line 194). Default-OFF + unmeasured on golden, not
new/unbuilt.",
    "severity": "medium"
},
{
    "file": "docs/MULTI_SIGNAL_FUSION_PLAN.md",
    "stale_text": "| `shuttle_player_colocation` | low (free space) | high (near a hand) |
**NEW** (needs the person cue) |",
    "corrected_text": "| `shuttle_player_colocation` | low (free space) | high (near a hand) |
**BUILT** (`fusion_features.py`; default-OFF under the person cue, unmeasured on golden) |",
    "why_stale": "Tags shuttle_player_colocation as \"NEW (needs the person cue)\" implying not
built. It IS built: backend/pipeline/detectors/fusion_features.py:168 (shuttle_player_colocation
function), included in PERSON_FEATURE_NAMES (line 197) and computed by compute_person_features
(line 213); wired in fusion_hybrid.py under cue_flags.person. It is default-OFF + mock-tested +
unmeasured on golden, not unbuilt.",
    "severity": "high"
},
{
    "file": "docs/MULTI_SIGNAL_FUSION_PLAN.md",
    "stale_text": "| `shuttle_player_colocation` | fraction of frames the shuttle sits
inside/adjacent to a person box (hand region) | **held = high; in-flight = mostly free space** ?
the direct held-shuttle discriminator |\n\nThese join the shuttle features in the candidate
`stats` JSON",
    "corrected_text": "| `shuttle_player_colocation` | fraction of frames the shuttle sits
inside/adjacent to a person box (hand region) | **held = high; in-flight = mostly free space** ?
the direct held-shuttle discriminator |\n\nThese five person features are now **built**
(`fusion_features.py`:PERSON_FEATURE_NAMES`; default-OFF, unmeasured on golden) and join the
shuttle features in the candidate `stats` JSON",
    "why_stale": "The heading above this table (\"Person features to add (~4, deliberately
few)\") and the prose frame these features as not-yet-built. All five (players_in_court,
two_sided_motion, mean_player_motion, in_court_ratio, shuttle_player_colocation) are built in
backend/pipeline/detectors/fusion_features.py (PERSON_FEATURE_NAMES, line 194) and persisted by
fusion_hybrid.py::_person_features. §10 P2 (line 278) already records them as shipped. They are
default-OFF + unmeasured on golden, not pending.",
    "severity": "low"
},
{
    "file": "docs/MULTI_SIGNAL_FUSION_PLAN.md",
    "stale_text": "| `net_crossings` | ? 1 (usually several) | ~0 (held) / ~1 (single feed) |
**EXISTS** (`rally_gate.py`); eval-only ? wire to prod |",
    "corrected_text": "| `net_crossings` | ? 1 (usually several) | ~0 (held) / ~1 (single feed)
| **EXISTS** (`rally_gate.py`); served Gate A wired via `FusionConfig.min_crossings` (default
0=OFF) |",
    "why_stale": "Status column still says Gate A is \"eval-only ? wire to prod\", but the
served wiring exists: FusionConfig.min_crossings (backend/config/models.py:105) is applied in
fusion_hybrid.py::_select_for_handoff (line 164) to drop sub-threshold windows from the served
handoff. The same doc's §1 (lines 60-64) already states \"P2 closed this:
FusionConfig.min_crossings now exists (default 0 = OFF)\", so this table row contradicts the
doc's own §1 and the code.",
    "severity": "medium"
}
]

```

2. assistant (2026-06-19T15:52:08.779Z)

I'll verify each proposed fix by checking the file text and the cited code. Let me start by reading the target doc and the relevant code files.

3. user (2026-06-19T15:52:09.677Z)

```
1      # Multi-signal rally fusion ? add a human (person) cue alongside the shuttle
2
3      **Status:** **P1 DONE** (2026-06-13) ? the person cue is extracted to
4      `backend/pipeline/detectors/person_detector.py` (`PersonDetector`), consumed by
5      `yolo_hybrid` with no
6      behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a
7      separate PR.
8      **Owner decisions captured** in [S9](#9-open-decisions-for-the-owner).
9
10     > **Companion docs:** the 2-layer mental model is
11     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md);
12     > the modular-fusion architecture this builds on is **ADR-007**
13     ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md));
14     > the permissive-dependency rule is **ADR-002 / ADR-005** ; the audio cue (the #1 next
15     cue, already
16     scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md).
17     > The vendor labels that gate the learned path come from
18     [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)
19     > and the labeling spec in [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md).
20     > **How each cue here is added/tested without production regression** ? A/B, shadow
21     eval, and the
22     > promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)
23     (every P1?P4 cue
24     > below lands flag-gated, default OFF, promoted only on measured LOVO lift).
25
26     ---
27
28     ## 1. Context ? why this, why now
29
30     Rally detection today runs **one cue at a time**. Each segmenter independently turns a
31     single signal
32     into rally windows:
33
34     | Segmenter | Signal | File |
35     |---|---|---|
36     | `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory)
37     | `backend/pipeline/detectors/tracknet_runner.py` |
38     | `yolo_hybrid` (the Pydantic `default_segmenter` fallback; `config.json` overrides it
39     to `gemini`) | person velocity (torchvision + ByteTrack) |
40     | `backend/pipeline/segmenters/yolo_hybrid.py` |
41     | `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |
42     | `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |
43
44     **There is no fusion.** The owner's directive: make the detector **use the shuttle *and*
45     the humans
46     together** ? and be ready to fold in more signals (audio, later pose) ? delivered
47     **MIT / Apache-2.0 / BSD-clean** so we can ramp commercially without a licensing
48     rethink.
49
50     This is **not greenfield** ? it operationalizes work the repo already blessed:
51
52     - **`HOW_RALLY_DETECTION_WORKS.md` open question:** "Fusion: combine shuttle-motion
53     (WASB) with
54     player-motion for static cams."
55     - **ADR-007** already designed a **modular "rally layer"** that fuses cues *outside* the
56     frozen WASB
57     model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks
58     up the parked
59     person cue and generalizes the fusion seam so future cues are "register a producer,"
60     not "re-touch
61     the segmenter."
62     - **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate,
```

```

explicitly
45     labelled in its own docstring as "Gate A in the over-fire fix spectrum (**B = player-
stance state
46     machine, future)"." The person cue in this plan **is that named-but-unbuilt Gate
B.**
47
48     ### The owner's "held-shuttle" false positive ? what's actually going on
49
50     The owner observed a "rally" that was just a player **holding / flicking the shuttle in
hand** between
51     points. This is the *exact* failure `rally_gate.py` was written to kill:
52
53     > "WASB over-detects: when a player flicks/tosses the shuttle back to the opponent for
service after
54     > losing a point, that single trajectory looks like a rally. The discriminator is
structural: a real
55     > rally has the shuttle crossing the net MULTIPLE times; a single service feed crosses
~once.""
56
57     **The structural fix already exists and is validated offline ? but it is NOT wired into
the production
58     runtime path.** `rally_gate.apply_gate(..., min_crossings=2)` is called only from the
**eval drivers**
59     (`eval_partial_labels.py`, `export_wasb_segments.py`, `calibrate_local.py`;
`distill_local.py` uses
60     `rally_gate.gate_windows` to compute the `net_crossings` feature, not `apply_gate`); the
live segmenters
61     historically didn't apply it. P2 closed this: `FusionConfig.min_crossings` now exists
(default `0` =
62     OFF), so the served Gate A is a config flag ? consistent with $10 P2 below, marked DONE.
So the
63     single cheapest, highest-confidence win in this whole plan is **P2's first step: fold
the existing
64     Gate A into production.** Everything else (the person cue / Gate B, the learned fusion)
is robustness
65     on top of that.
66
67     ### Why humans help ? and the honest risk
68
69     The beachhead videos are **static tripod** (Bangalore academies,
[PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md)),
70     so player presence/motion is a usable cue. ADR-001's warning was specifically about
**broadcast pan**
71     cameras, where player-motion fooled the old heuristic ? *that failure mode does not
apply to a fixed
72     court cam.* But we keep **the shuttle as the camera-invariant primary** so the general
bet stays robust
73     on other camera types. Humans then add:
74
75     - **Precision** ? reject shuttle-on-floor / warm-up knock-ups / held-shuttle feeds when
the court
76     isn't occupied by ?2 players in a play posture.
77     - **Recall** (via the learned classifier) ? features that help bridge shuttle
occlusion/blur gaps.
78
79     ---
80
81     ## 2. Core architecture ? one rally-fusion layer, cues are swappable producers
82
83     Formalize ADR-007's diagram. Every cue is a **black box that emits a time-aligned
signal** (per-frame
84     activity and/or per-window features); **fusion happens in *our* layer, never inside a
frozen model.**
85
86     ...

```

```

87     frames ?? WASB (frozen, MIT)      ?? shuttle trajectory ???
88     frames ?? person detector (ours) ?? tracks + boxes ???????
89     audio  ?? hit detector (ours)     ?? hit events ???????????? RALLY LAYER (ours) ??
rallies
90     [frames ?? pose/stance (ours; parked #3) ?? stance ???????? feature + decision fusion
91     ```
92
93     - **Shuttle = primary, camera-invariant.** Windows still **seed** from
94     `TrackNetRunner.trajectory_to_action_windows()` (`tracknet_runner.py:234`). Other cues
*enrich and
95     adjudicate* shuttle-seeded windows; they do **not** replace the shuttle seed. Person
runs only over
96     **shuttle-seeded candidate windows + padding**, not the whole video ? which bounds the
added cost.
97     - **A small `RallyCue` contract.** Each producer yields `(frame_idx ? features)` + per-
window stats.
98     Adding a cue = register a producer; the existing `SegmenterRegistry`
(`segmenters/base.py:35`) is the
99     template for the pattern. Keep the single-cue segmenters registered as fallbacks.
100
101     ---
102
103     ## 3. The two fusion modes (both ? per the owner decision)
104
105     ### 3.1 Decision-level gate (safe baseline, ships first, no labels needed)
106
107     ```
108     rally_active = shuttle_motion ? (?? persons in court zone ? recent person track)
109     ```
110
111     The `? recent-track` clause tolerates 1?2 frame person-detector misses so the AND-gate
can't cause a
112     recall cliff. This composes with the **existing net-crossing Gate A**:
113
114     ```
115     keep window ? net_crossings ? 2                (Gate A ? EXISTS in rally_gate.py, eval-
only today)
116                ? (?? in-court players ? recent track)  (Gate B ? the person cue, NEW)
117     ```
118
119     Gate A kills the service-feed / held-shuttle case structurally; Gate B adds court-
occupancy evidence
120     for the cases a single trajectory can still fool (e.g. a knock-up that does cross the
net). No training
121     data required ? this is the label-free fusion that ships before any golden labels exist.
122
123     ### 3.2 Feature-level (primary, the learned path)
124
125     The cues emit a **small set of aggregated, scale/translation-invariant per-window
features ? NOT raw
126     coordinates.** Raw pixel coordinates would memorize *this* court/camera and fail to
generalize off a
127     still-small labeled set; counts / ratios / zone-membership / two-sidedness transfer.
128
129     **What already feeds the classifier** (`distill_local.py:40`, `FEATURE_NAMES`):
130     `duration, peak_velocity, mean_velocity, active_ratio, net_crossings` ? note
**`net_crossings` is
131     already a feature** (computed via `rally_gate.gate_windows`).
132
133     **Person features to add (~4, deliberately few):**
134
135     | Feature | Meaning | Rally signal |
136     |---|---|---|
137     | `players_in_court` | median count of persistent in-court tracks | ? 2 (singles) / 4
(doubles); 0?1 = dead time |

```

```

138 | `two_sided_motion` | are moving players on both net-halves? | rally = two-sided;
one person feeding = one-sided |
139 | `mean_player_motion` | aggregate normalized player velocity | rally = sustained;
resting/standing = low |
140 | `in_court_ratio` | fraction of frames with ?2 in-court players | track stability vs
flicker |
141 | `shuttle_player_colocation` | fraction of frames the shuttle sits inside/adjacent to a
person box (hand region) | held = high; in-flight = mostly free space ? the direct held-
shuttle discriminator |
142
143 These join the shuttle features in the candidate `stats` JSON
144 (`database.py::add_candidate_segment`) and the feature vector of the existing
distilled classifier ?
145 the tiny numpy-only L2 logistic regression in `backend/eval/classifier.py` (ADR-004),
NOT a new net.**
146 It scores each candidate window `P(real rally)` and replaces the Gemini call at runtime.
147
148 What the model learns: the weights on those few features ? e.g. down-weight a
shuttle-motion
149 window with 0?1 in-court players / one-sided motion / high shuttle-colocation (warm-up,
floor shuttle,
150 held-shuttle feed); up-weight ?2 players, two-sided, sustained motion, shuttle in free
flight.
151
152 Small-N discipline (non-negotiable):
153 - Keep to ~4 person features ? more features on a still-small labeled set overfit;
that's exactly
154 why the model stays a logistic regression, not a net.
155 - Use the leave-one-video-out (LOVO) protocol that `distill_local.py` already
implements**: train
156 on the other videos' candidates, predict the held-out video, score vs its labels ?
never a
157 held-out window from the same video (windows in one video are correlated; that
flatters the score).
158 - Training labels today are Gemini silver labels (distilled offline, ADR-004/007 ?
Gemini is a
159 teacher, never a runtime dependency, never a training-data source for owned
weights); golden
160 labels are the honest measuring stick**, and as the Roora golden corpus grows they can
also become
161 training labels. Until the labeled set is larger, treat the learned weights as
162 calibration / direction-check, not "a fully general model" ? the decision-level
gate (§3.1) is
163 the shippable, label-free fusion in the meantime.
164
165 > *(Gated on golden labels existing ? the same blocker ADR-007 flags for audio. See
166 > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md).)*
167
168 ---
169
170 ## 4. Shuttle-state: in-flight vs in-hand (the misclassification, mapped to real code)
171
172 The "shuttle moved" seed conflates a shuttle in flight with a shuttle carried in
hand. The
173 discriminators, and where each lives:
174
175 | Signal | In-flight | Held / static | Status |
176 |---|---|---|---|
177 | `net_crossings` | ? 1 (usually several) | ~0 (held) / ~1 (single feed) | EXISTS
(`rally_gate.py`); eval-only ? wire to prod |
178 | `shuttle_player_colocation` | low (free space) | high (near a hand) | NEW (needs
the person cue) |
179 | `direction_reversals` | several (racket contacts) | ~0 | derivable from trajectory;
partial overlap with crossings |
180 | `peak shuttle speed` | high | walking speed | derivable from existing velocity |

```

```

181
182     So the in-flight-vs-held fix is **mostly Gate A (already built) + the new colocation
feature from the
183     person cue**. **No new label column is needed** ? the golden labels already start a
rally at the
184     *serve-contact frame* and treat all pre-serve / held-shuttle time as dead time (see
185     [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md)), so held-shuttle moments
are negatives
186     by construction and are the ground truth that proves the detector wrong.
187
188     ---
189
190     ## 5. Adversarial mitigations (integrating safely)
191
192     - **Court-region mask** ? polygon around the court; drop persons outside it (spectators,
coaches,
193     adjacent courts). Config-supplied per camera; ~1 ms. Source: a manual per-camera
polygon first;
194     the **4 court corners + net-line** that Roora labels once per video (see the labeling
doc) build
195     this mask **and** the net split that `two_sided_motion` / `net_crossings` need.
196     - **Player-count constraint** ? expect **2 (singles) / 4 (doubles)** persistent in-court
tracks;
197     suppress transient/extra detections. The `singles_or_doubles` manifest flag sets the
expectation.
198     - **Temporal association** ? reuse `supervision` **ByteTrack** (already in
`yolo_hybrid`); reject
199     singleton detections; smooth across frames.
200     - **Multi-court rejection** ? proximity to the primary court region (homography
optional, later).
201     - **Domain shift (COCO frontal ? side/overhead court)** ? start **box-only** (robust);
keep the
202     detector swappable. Fine-tuning a person detector is **parked** ? we have no in-house
person labels,
203     and the **paid-LLM/training guardrail + minors-exclusion** apply to any future person-
training data.
204     - **Fusion-degradation guard** ? shuttle stays primary; person is gate + feature, never
the sole
205     trigger; the single-cue segmenters remain registered fallbacks.
206
207     ---
208
209     ## 6. Box vs pose
210
211     **Box-first.** Occupancy + count + two-sided motion + shuttle-colocation is enough for
both the gate
212     and the features, and box detection is the cheap, robust win. **2D pose is parked behind
the same
213     `RallyCue` interface** ? pose (MediaPipe Pose / RTMPose, both Apache-2.0) buys serve-
ready stance
214     gating and shot cues later, at the cost of a second model. This matches ADR-007 parking
pose as the
215     #2/#3 cue.
216
217     ---
218
219     ## 7. Performance
220
221     The person detector is the **existing in-process torchvision model** run at **frame-skip
375375
222     (`yolo_frame_skip` default 3) with track interpolation between sampled frames. The
shuttle keeps
223     running in its separate WSL process ? the two are **separate processes**, so a "shared
backbone" isn't
224     free; budget realistically as the existing `yolo_hybrid` cost minus skip savings. Person

```

```

runs **only
225     over shuttle-seeded windows + `ai_handoff_padding`, not the whole video, which bounds
the cost.
226
227     ---
228
229     ## 8. Licensing (the guardrail table ? all green)
230
231     Every dependency stays **MIT / Apache-2.0 / BSD** (ADR-002 / ADR-005; `ultralitics` was
dropped for
232     AGPL-3.0).
233
234     | Cue / model | Code | Weights / data | Verdict |
235     |---|---|---|---|
236     | Shuttle: WASB-SBDT, TrackNetV3 | MIT | frozen badminton weights | ? (ADR-001/002) |
237     | Person (default): torchvision Faster R-CNN / RetinaNet / FCOS | BSD/MIT | COCO-
pretrained | ? already in repo (`_DETECTOR_FACTORIES`) |
238     | Person (escalation): **YOLOX**, **RT-DETR / RT-DETRv2** | Apache-2.0 | COCO /
Objects365 | ? |
239     | Pose (parked): **MediaPipe Pose**, **RTMPose / MMPose** | Apache-2.0 | COCO-structure
| ? |
240     | Tracking: `supervision` ByteTrack | MIT | n/a | ? already in repo |
241     | **Banned** | **Ultralytics YOLOv5/8/11 (AGPL-3.0)** . **MonoTrack (Adobe, non-comm)**
. **YOLO-NAS (Deci, non-comm)** | ? | ? |
242
243     **COCO caveat to record:** COCO *annotations* are **CC BY 4.0** (commercial OK *with
attribution*);
244     COCO *images* are Flickr-ToS (per-image). Pretrained-weight use is clean; record the
attribution.
245
246     ---
247
248     ## 9. Open decisions (for the owner)
249
250     1. Court-region mask source ? manual per-camera polygon (simple, ship first) vs auto
court
251         homography/line detection (later). Recommend manual first, fed by Roora's court-
corner labels.
252     2. Person detector default ? keep torchvision (zero new deps, BSD/MIT) vs adopt
YOLOX/RT-DETR
253         (Apache, faster) now. Recommend torchvision first, escalate only if eval shows a
speed/accuracy gap.
254     3. Pose now or parked ? recommend parked behind the interface (box-first).
255     4. Golden labels ? feature-level fusion (P3) is blocked on golden rally labels;
the source is
256         the VLC `rally-annotator` + the Roora vendor pilot
([GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)).
257
258     ---
259
260     ## 10. Phased execution (owner-gated; ONE PR per slice, off `master`)
261
262     - P0 ? this design doc +
[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md), indexed in
263     `docs/README.md`, cross-linked from `HOW_RALLY_DETECTION_WORKS.md` and ADR-007. *(this
deliverable;
264     docs-only, no code.)*
265     - P1 ? extract the shared person cue: DONE (2026-06-13). Lifted the
torchvision detector +
266     ByteTrack + velocity out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
267     (`PersonDetector`); `yolo_hybrid` now orchestrates only and delegates to `self.person`
(no behavior
268     change ? regression-safe, proven by the unchanged process_video output tests). New
269     `tests/test_person_detector.py`; suite +1 green; person_detector is type-clean (left

```



```

mypy quarantine
270     for the moved code). The fusion segmenter (P2) reuses this same producer over shuttle-
seeded windows.
271     - **P2 ? the FusionSegmenter (Gate A + Gate B + person-feature persistence):** ? **DONE
(2026-06-14).**
272     Shipped `backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter` (registry
`fusion_hybrid`,
273     **default-OFF** ? `default_segmenter` unchanged), subclassing the trajectory engine
via two identity
274     hooks (`_enrich_candidate_stats`, `_select_for_handoff` ? wasb/tracknet stay byte-
identical, adversarially
275     verified). It seeds windows from the shuttle substrate, **always persists
`net_crossings`** (Gate A signal),
276     and ? when `indexing.fusion.cue_flags.person` is on ? runs the person cue over each
window's frames
277     (`PersonDetector.detections_per_frame`) and persists the ~5 person features
(`fusion_features.py`,
278     pure + unit-tested). Served gating is optional + default-OFF: `fusion.min_crossings`
(Gate A) /
279     `fusion.min_players` (Gate B) skip sub-threshold windows from the handoff while
**persisting the full
280     superset** (offline substrate intact). Court mask = optional `fusion.court_polygon`
(None ? count-only).
281     Lift to be measured via the harness; held-shuttle-drop + person-feature values are
GPU/owner-side.
282     - **P3 ? feature-level (learned) fusion:** add the person features (incl.
`shuttle_player_colocation`)
283     to `distill_local` / `classifier`; recalibrate IoU on the golden set; report **LOVO
lift on held-out
284     videos**. *(Gated on golden labels.)*
285     - **P4 ? additional cues:** wire **audio** (ADR-007's #1, already scaffolded in
286     `backend/pipeline/audio/hit_detector.py`) and later **pose**, through the same
`RallyCue` interface.
287     Each is add-a-producer, not a refactor.
288
289     ---
290
291     ## 11. Verification
292
293     - **Offline, runnable in this dev container (no GPU/torch/cv2):** unit tests for the
fusion gate logic
294     and `RallyCue` aggregation (court-mask filter, player-count constraint, `? recent-
track` tolerance)
295     and the shuttle-state signals (net-crossing count ? extends the existing `rally_gate`
tests ? and
296     person-colocation) with **synthetic trajectories + person boxes**. A key-license
assertion test
297     (no `ultralytics` import; detector factories stay torchvision/Apache).
298     - **Machine-side (owner, GPU/WSL ? the full pipeline can't run in this container):** run
the
299     `FusionSegmenter` on a real static-tripod clip; confirm **held-shuttle / warm-up /
spectator false
300     positives drop** vs shuttle-only (this should reproduce the owner's observed bug and
show it fixed ?
301     start by confirming Gate-A-in-production alone fixes it); **LOVO IoU lift on the
golden set** for the
302     learned path.
303     - **Docs (P0):** links resolve; both docs indexed in `docs/README.md`; this doc cross-
linked from
304     `HOW_RALLY_DETECTION_WORKS.md` (the fusion open-question) and ADR-007.
305

```

4. user (2026-06-19T15:52:10.606Z)

```

1     """Multi-signal fusion segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

```

```

2
3 Shuttle stays PRIMARY: this reuses an existing trajectory substrate (WASB by default,
4 TrackNet optional) to seed candidate windows exactly as `wasb_hybrid`/`tracknet_hybrid`
5 do ? it subclasses the same `TrajectoryHybridSegmenter` engine. It then ENRICHES each
6 shuttle-seeded window via two overridable seams the engine exposes:
7
8 - ``_enrich_candidate_stats`` ? always adds the **net-crossing count** (Gate A signal,
9 GPU-free, computed from the trajectory we already have); and, only when the person cue
10 is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
11 (``fusion_features``), persisting them into ``candidate_segments.stats`` for the learned
12 fusion (P3). The person path lazily builds ONE `PersonDetector` and only then loads
the
13 torchvision/supervision model ? so the default (person-OFF) path never loads a
detector
14 model nor touches the GPU. (torch/cv2 themselves are imported by the registry
regardless,
15 via the pre-existing yolo_hybrid ? person_detector chain; the lazy part is the model.)
16 - ``_select_for_handoff`` ? optional served Gate A/B: when
``indexing.fusion.min_crossings``
17 (and, with the person cue, ``min_players``) are raised above 0, sub-threshold windows
are
18 dropped from the AI handoff (the held-shuttle / one-sided-feed false-positive fix).
19 **Persisted candidates remain the full superset** regardless, so the offline
experiment
20 harness can re-score any variant.
21
22 Defaults reproduce the substrate exactly: net_crossings/person features are persisted
but
23 nothing is gated (thresholds 0, person OFF), so `FusionSegmenter` with default config
seeds
24 and hands off identically to `wasb_hybrid`. Registered as ``fusion_hybrid``, NOT the
default
25 segmenter ? it is opt-in. Promotion to default happens only on measured LOVO lift
through the
26 experiment harness (EXPERIMENT_HARNESS.md), never silently.
27 """
28 from typing import Any, Dict, List, Optional, Tuple
29
30 from backend.pipeline.segmenters.base import segmenter_registry
31 from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
32 from backend.pipeline.detectors.base import DetectorRunner
33 from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
34 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
35 from backend.pipeline.detectors import rally_gate
36 from backend.pipeline.detectors import fusion_features as ff
37
38
39 class FusionSegmenter(TrajectoryHybridSegmenter):
40     """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate
B)."""
41
42     family = "fusion"
43     display_label = "Fusion"
44
45     def __init__(self, db: Any, config: Any):
46         super().__init__(db, config)
47         # Built lazily only if the person cue is enabled (no detector model loaded
otherwise).
48         self._person: Optional[Any] = None
49         # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-
estimate it
50         # over the whole trajectory once per candidate window (it depends only on
`points`).
51         self._axis_cache_key: Optional[int] = None
52         self._axis_cache: Optional[tuple] = None

```

```

53
54     @property
55     def name(self) -> str:
56         return "fusion_hybrid"
57
58     @property
59     def description(self) -> str:
60         return ("Multi-signal fusion: shuttle trajectory (WASB/TrackNet) seeds candidate
rallies, "
61                 "net-crossing Gate A + optional person-occupancy Gate B adjudicate them,
and the "
62                 "AI provider sets semantic boundaries.")
63
64     def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
65         substrate = idx_cfg.get("fusion", {}).get("substrate", "wasb")
66         if substrate == "tracknet":
67             cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
68             return TrackNetRunner(cfg), {
69                 "weights_path": cfg.weights_path,
70                 "queue_length": cfg.queue_length,
71                 "fusion_substrate": "tracknet",
72             }
73         wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
74         return WasbRunner(wcfg), {"weights_path": wcfg.weights_path, "fusion_substrate":
"wasb"}
75
76     def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
77                                frame_width: float, video_path: str,
78                                idx_cfg: Dict[str, Any]) -> Dict[str, Any]:
79         stats = super()._enrich_candidate_stats(cw, points, fps, frame_width,
video_path, idx_cfg)
80         # Gate-A signal ? net-crossing count for this window. Pure, GPU-free, always
persisted
81         # (single service feed crosses ~once; a real rally crosses >=2x). axis/net auto-
estimated
82         # over the whole trajectory by rally_gate (camera-agnostic); reused across
windows.
83         gated = rally_gate.gate_windows(points, [(cw["start"], cw["end"])], fps,
84                                               min_crossings=0,
estimate=self._axis_estimate(points))
85         stats["net_crossings"] = float(gated[0].crossings) if gated else 0.0
86
87         fcfg = idx_cfg.get("fusion", {})
88         if fcfg.get("cue_flags", {}).get("person", False):
89             stats.update(self._person_features(cw, points, fps, frame_width, video_path,
fcfg))
90         return stats
91
92         # P3 (learned fusion) first-step landing note (per MULTI_SIGNAL_FUSION_PLAN +
NEXT_STEPS 2026-06-14):
93         # Person features (from _person_features when cue_flags.person) are now persisted
for the harness
94         # `compare` LOVO litmus on the 6 golden videos. Eager-person-compute optimisation
(compute only
95         # for shuttle-seeded windows + padding, not whole video) is already in place via the
lazy
96         # self._person + windowed detections_per_frame. Next: actual harness run + lift
measurement
97         # (owner/hardware) before any served promotion. Flag-gated (default person OFF). See
98         # docs/NEXT_STEPS.md "Rally-detection quality track" and EXPERIMENT_HARNESS.md for
discipline.
99
100     def _axis_estimate(self, points: List[Any]) -> tuple:
101         """Cache rally_gate's (axis, net, span) play-axis estimate per trajectory so

```

```

it's
102         computed once per video, not once per candidate window (it depends only on
points)."""
103         key = id(points)
104         if self._axis_cache_key != key or self._axis_cache is None:
105             self._axis_cache_key = key
106             self._axis_cache = rally_gate.estimate_play_axis(points)
107         est = self._axis_cache
108         assert est is not None
109         return est
110
111     def _person_features(self, cw: Dict[str, Any], points: List[Any], fps: float,
112                         frame_width: float, video_path: str,
113                         fcfg: Dict[str, Any]) -> Dict[str, float]:
114         """GPU path (person cue ON): run the person detector over this window's
ORIGINAL-video
115         frame range and compute the scale/translation-invariant person features. Lazily
builds
116         ONE PersonDetector. ? real-video verification is owner-side; this wiring is
mock-tested."""
117         # Lazy import so the default (person-OFF) path never pulls torch/cv2 through
this module.
118         from backend.pipeline.detectors.person_detector import PersonDetector
119         if self._person is None:
120             self._person = PersonDetector(self.config)
121
122         frame_skip = self.config.get("indexing", {}).get("yolo_frame_skip", 3)
123         # round (not truncate): int() would map e.g. 1.999s*30 to frame 59 not 60 ? a
124         # window-edge off-by-one that drops the boundary frame from the shuttle
alignment.
125         start_f, end_f = round(cw["start"] * fps), round(cw["end"] * fps)
126         det = self._person.detections_per_frame(video_path, start_f, end_f, frame_skip)
127         fw = det["frame_width"] or frame_width
128         fh = det["frame_height"]
129         # If we can't trust the frame dims, every normalised feature would silently
collapse
130         # to ~0 ? return an explicit all-zero person-feature row instead of a half-baked
one.
131         if fw <= 0 or fh <= 0:
132             return {k: 0.0 for k in ff.PERSON_FEATURE_NAMES}
133
134         poly = fcfg.get("court_polygon")
135         polygon = [(float(p[0]), float(p[1])) for p in poly] if poly else None
136         net = (fcfg.get("net_axis", "y"), float(fcfg.get("net_position", 0.5)))
137         shuttle_in = [p for p in points if start_f <= p.frame <= end_f]
138         return ff.compute_person_features(det["frames"], shuttle_in, fw, fh,
139                                         court_polygon=polygon, net=net)
140
141     def _select_for_handoff(self, windows: List[Dict[str, Any]],
142                           window_stats: List[Dict[str, Any]],
143                           idx_cfg: Dict[str, Any]) -> List[int]:
144         """Served Gate A/B over the (already-enriched) per-window stats. Default
thresholds
145         (0) keep every window ? identical to the substrate. Persisted candidates are
untouched (full superset).
146
147         NEVER-EMPTY safeguard: a video that had ?1 window must keep ?1 after gating. If
the
148         thresholds would drop EVERY window (e.g. a short, noisy or one-sided-feed
trajectory
149         where no segment clears ``min_crossings``), the gate falls back to the full
ungated
150
151         set rather than emit a video with zero rallies. This makes Gate A/B strictly
152         *pruning* ? it can shrink a candidate set but never zero it out ? so layering it
on

```

```

153         the substrate is "do no harm": the bounded-windowing / largest-internal-gap
split
154         below it always wins on an otherwise-empty video. Observed on real footage
155         (mahadevpura-1): a single mega-window split into pieces, none clearing the
crossings
156         gate ? without this guard the whole clip yielded 0 rallies (no reel)."""
157         fcfg = idx_cfg.get("fusion", {})
158         min_crossings = int(fcfg.get("min_crossings", 0))
159         min_players = int(fcfg.get("min_players", 0))
160         person_on = bool(fcfg.get("cue_flags", {}).get("person", False))
161
162         keep: List[int] = []
163         for i, st in enumerate(window_stats):
164             if min_crossings > 0 and st.get("net_crossings", 0.0) < min_crossings:
165                 continue
166             if person_on and min_players > 0 and st.get("players_in_court", 0.0) <
min_players:
167                 continue
168             keep.append(i)
169
170         if not keep and window_stats:
171             # Gating emptied a non-empty video ? fall back to the ungated set.
172             return list(range(len(window_stats)))
173         return keep
174
175
176     segmenter_registry.register(FusionSegmenter)
177

```

5. user (2026-06-19T15:52:11.093Z)

```

1     """Pure per-window person-cue features for multi-signal rally fusion
(MULTI_SIGNAL_FUSION_PLAN §3.2/§4).
2
3     NO I/O, no torch, no cv2 ? mirrors `rally_gate.py`'s discipline so the whole feature
4     math is unit-testable in a GPU-free dev container. The detection (person boxes,
5     shuttle trajectory) is produced elsewhere (`PersonDetector`, the WASB runner); this
6     module only turns those signals into a handful of scale/translation-invariant
7     features (counts / ratios / zone-membership ? never raw pixel coordinates, which would
8     memorise this court/camera and fail to generalise off a small labelled set).
9
10    Conventions:
11    - A person box is `(x1, y1, x2, y2)` in pixels; `(track_id, box)` when motion
needs it.
12    - `per_frame` is `{frame_idx: [(track_id, box), ...]}` keyed by the ORIGINAL video
frame
13        index (so it aligns directly with the shuttle trajectory's `frame` ? no clip re-
timing).
14    - A shuttle point has `frame`, `visible`, `x`, `y` (pixels) ? the WASB
trajectory shape.
15    - `court_polygon` is normalised 0-1 `[(x, y), ...]` or `None` (None ? no mask,
every
16        detection counts ? the player-count-only mode).
17    - `net` is `(axis 'x'|'y', position 0-1 normalised)` or `None` (None ? two-sided =
0.0).
18
19    Every function degrades gracefully (returns 0.0 / empty) on missing inputs; outputs are
20    floats in [0, 1] except `mean_player_motion` (a non-negative normalised speed).
21    """
22    from __future__ import annotations
23
24    from typing import Dict, List, Optional, Sequence, Tuple
25
26    BBox = Tuple[float, float, float, float]
27    Net = Tuple[str, float]

```

```

28
29
30 def point_in_polygon(point: Tuple[float, float], polygon: Sequence[Tuple[float, float]])
-> bool:
31     """Ray-casting point-in-polygon. `point` and `polygon` share a coordinate space
32     (we use normalised 0-1). Empty/degenerate polygon (<3 vertices) ? False."""
33     if polygon is None or len(polygon) < 3:
34         return False
35     x, y = point
36     inside = False
37     n = len(polygon)
38     j = n - 1
39     for i in range(n):
40         xi, yi = polygon[i]
41         xj, yj = polygon[j]
42         # Does the horizontal ray at y cross edge (i, j)?
43         if (yi > y) != (yj > y):
44             x_cross = (xj - xi) * (y - yi) / (yj - yi) + xi if yj != yi else xi
45             if x < x_cross:
46                 inside = not inside
47         j = i
48     return inside
49
50
51 def _foot_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float,
float]:
52     """Normalised foot point (bottom-centre) of a person box ? where the player stands,
53     the right anchor for court membership. Returns (0,0) if frame dims are unusable."""
54     if frame_width <= 0 or frame_height <= 0:
55         return (0.0, 0.0)
56     x1, _y1, x2, y2 = box
57     return ((x1 + x2) / 2.0 / frame_width, y2 / frame_height)
58
59
60 def _center_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float,
float]:
61     if frame_width <= 0 or frame_height <= 0:
62         return (0.0, 0.0)
63     x1, y1, x2, y2 = box
64     return ((x1 + x2) / 2.0 / frame_width, (y1 + y2) / 2.0 / frame_height)
65
66
67 def person_in_court(box: BBox, frame_width: float, frame_height: float,
68                     court_polygon: Optional[Sequence[Tuple[float, float]]]) -> bool:
69     """Is this person on the court? Foot-point-in-polygon when a polygon is supplied;
70     True (count everyone) when it is None. Unusable frame dims (w/h <= 0) make the foot
71     point meaningless, so a masked check returns False rather than testing a bogus
(0,0)."""
72     if court_polygon is None:
73         return True
74     if frame_width <= 0 or frame_height <= 0:
75         return False
76     return point_in_polygon(_foot_norm(box, frame_width, frame_height), court_polygon)
77
78
79 def in_court_counts(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
80                     frame_height: float,
81                     court_polygon: Optional[Sequence[Tuple[float, float]]]) ->
List[int]:
82     """Per-frame count of in-court persons (one entry per sampled frame)."""
83     return [
84         sum(1 for _tid, box in dets if person_in_court(box, frame_width, frame_height,
court_polygon))
85         for _frame_idx, dets in sorted(per_frame.items())
86     ]

```

```

87
88
89 def _median(vals: Sequence[float]) -> float:
90     if not vals:
91         return 0.0
92     s = sorted(vals)
93     n = len(s)
94     return float(s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2]))
95
96
97 def players_in_court(counts: Sequence[int]) -> float:
98     """Median per-frame in-court count (?2 singles / 4 doubles; 0-1 = dead time)."""
99     return _median(list(counts))
100
101
102 def in_court_ratio(counts: Sequence[int], min_players: int = 2) -> float:
103     """Fraction of sampled frames with >= `min_players` in court (track stability vs
104 flicker)."""
105     if not counts:
106         return 0.0
107     return sum(1 for c in counts if c >= min_players) / len(counts)
108
109
110 def mean_player_motion(per_frame: Dict[int, List[Tuple[int, BBox]]],
111                        frame_width: float, frame_height: float) -> float:
112     """Mean per-track centre displacement between consecutive sampled frames, with x
113 normalised by frame width and y by frame height (per-axis; uniform-scale-invariant,
114 not aspect-ratio-isotropic). 0.0 if <2 frames or no shared tracks."""
115     if frame_width <= 0 or len(per_frame) < 2:
116         return 0.0
117     frames = sorted(per_frame.keys())
118     steps: List[float] = []
119     for a, b in zip(frames, frames[1:]):
120         prev = {tid: box for tid, box in per_frame[a]}
121         for tid, box in per_frame[b]:
122             if tid in prev:
123                 cx0, cy0 = _center_norm(prev[tid], frame_width, frame_height)
124                 cx1, cy1 = _center_norm(box, frame_width, frame_height)
125                 steps.append(((cx1 - cx0) ** 2 + (cy1 - cy0) ** 2) ** 0.5)
126     if not steps:
127         return 0.0
128     return sum(steps) / len(steps)
129
130
131 def two_sided_motion(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
132                     frame_height: float, net: Optional[Net],
133                     court_polygon: Optional[Sequence[Tuple[float, float]]]) -> float:
134     """Fraction of sampled frames with >=1 in-court person on BOTH net halves.
135
136     A real rally is two-sided; a single player feeding/holding the shuttle is one-sided.
137     The net splits the play axis at `net=(axis, position)` in normalised coords. None ?
138     0.0.
139
140     """
141     if net is None or not per_frame:
142         return 0.0
143     axis, pos = net
144     both = 0
145     for _frame_idx, dets in per_frame.items():
146         near, far = False, False
147         for _tid, box in dets:
148             if not person_in_court(box, frame_width, frame_height, court_polygon):
149                 continue
150             fx, fy = _foot_norm(box, frame_width, frame_height)
151             coord = fx if axis == "x" else fy
152             if coord < pos:

```

```

150         near = True
151     else:
152         far = True
153     if near and far:
154         both += 1
155     return both / len(per_frame)
156
157
158 def _shuttle_in_box(sx: float, sy: float, box: BBox, frame_width: float, frame_height:
float,
159                    pad_frac: float) -> bool:
160     """Is the (pixel) shuttle point inside a person box, expanded by `pad_frac` of frame
161     width/height (the 'adjacent / in-hand' tolerance)?"""
162     x1, y1, x2, y2 = box
163     px = pad_frac * frame_width
164     py = pad_frac * frame_height
165     return (x1 - px) <= sx <= (x2 + px) and (y1 - py) <= sy <= (y2 + py)
166
167
168 def shuttle_player_colocation(per_frame: Dict[int, List[Tuple[int, BBox]]],
shuttle_points,
169                               frame_width: float, frame_height: float,
170                               pad_frac: float = 0.03) -> float:
171     """Fraction of (visible shuttle frames that were also person-sampled) in which the
172     shuttle sits inside/adjacent to a person box. **held/in-hand ? high; in-flight ?
~0**
173     ? the direct held-shuttle discriminator. 0.0 if no aligned visible shuttle frames.
174     """
175     if not per_frame or frame_width <= 0 or frame_height <= 0:
176         return 0.0
177     aligned = 0
178     colocated = 0
179     for p in shuttle_points or []:
180         if not getattr(p, "visible", False):
181             continue
182         dets = per_frame.get(int(p.frame))
183         if dets is None:          # shuttle frame wasn't person-sampled ? can't judge
184             continue
185         aligned += 1
186         if any(_shuttle_in_box(p.x, p.y, box, frame_width, frame_height, pad_frac)
187               for _tid, box in dets):
188             colocated += 1
189     return colocated / aligned if aligned else 0.0
190
191
192 # The ~5 person features persisted into candidate `stats` (order = the learned-fusion
193 # feature columns appended after the shuttle features in distill_local.FEATURE_NAMES).
194 PERSON_FEATURE_NAMES = [
195     "players_in_court", "two_sided_motion", "mean_player_motion",
196     "in_court_ratio", "shuttle_player_colocation",
197 ]
198
199
200 def compute_person_features(per_frame: Dict[int, List[Tuple[int, BBox]]],
shuttle_points,
201                             frame_width: float, frame_height: float,
202                             court_polygon: Optional[Sequence[Tuple[float, float]]] =
None,
203                             net: Optional[Net] = None,
204                             min_players: int = 2) -> Dict[str, float]:
205     """Aggregate the per-window person features from per-frame detections + the in-
window
206     shuttle trajectory. Pure; returns exactly `PERSON_FEATURE_NAMES`, all floats."""
207     counts = in_court_counts(per_frame, frame_width, frame_height, court_polygon)
208     return {

```



```

209         "players_in_court": players_in_court(counts),
210         "two_sided_motion": two_sided_motion(per_frame, frame_width, frame_height, net,
court_polygon),
211         "mean_player_motion": mean_player_motion(per_frame, frame_width, frame_height),
212         "in_court_ratio": in_court_ratio(counts, min_players),
213         "shuttle_player_colocation": shuttle_player_colocation(
214             per_frame, shuttle_points, frame_width, frame_height),
215     }
216

```

6. user (2026-06-19T15:52:11.505Z)

```

1     """Pydantic models for the application configuration.
2
3     This module defines the canonical, validated, typed configuration for the
4     Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6     The models support graceful migration from old config.json files via extra="allow".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
Field(default_factory=ValidationRelevanceConfig)
31
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
43         velocity_threshold: float = 0.02
44         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
45         # decoded frame cache) are deleted automatically ? they are pure intermediates,
46         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
47         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
48         keep_frames: bool = False
49         # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB.
50         # 0 = disabled.
51         min_free_gb: float = 0.0
52

```

```

53
54 class WasbIndexConfig(BaseModel):
55     """Typed config for the live WASB shuttle substrate (indexing.wasb).
56
57     Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
58     these knobs actually take effect ? previously this whole block was silently dropped
59     because nested config sections defaulted to extra='ignore'.
60     """
61     wsl_distro: str = "Ubuntu"
62     repo_dir: str = "~/models/WASB-SBDT"
63     conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
64     conda_env: str = "wasb"
65     weights_path: str = "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
66     sport: str = "badminton"
67     wsl_stage_dir: str = "~/clips"
68     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
69     velocity_threshold: float = 0.02
70     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
71     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
72     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
73     keep_frames: bool = False
74     # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 =
disabled.
75     min_free_gb: float = 0.0
76     # FAST PATH (default OFF until owner-verified side-by-side): decode the staged video
77     # on the fly and feed frames straight to the detector, skipping the slow per-frame
PNG
78     # extraction that dominates a long clip (~46k PNGs for a 12-min 1080p60 video, which
79     # overran the 30-min timeout). Output is bit-identical to the PNG path (PNG is
lossless),
80     # so this is purely a speed/disk win; kept opt-in until the side-by-side confirms
it.
81     stream_video: bool = False
82
83
84 class FusionConfig(BaseModel):
85     """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
86
87     Every default reproduces control behaviour: the fusion segmenter persists the
88     shuttle net-crossing count (Gate A signal) and ? only when `cue_flags['person']`
is
89     true ? the person-cue features, but it does NOT gate the served handoff unless a
90     threshold is raised (`min_crossings` / `min_players` default 0 = OFF). The court
mask
91     is optional: `court_polygon=None` ? Gate B counts every detection (player-count-
only);
92     supplied ? off-court persons (spectators / adjacent courts) are excluded.
93     """
94     # Which trajectory substrate seeds the windows (shuttle stays primary).
95     substrate: Literal["wasb", "tracknet"] = "wasb"
96     # Windowing velocity threshold for the fusion family (the engine reads
97     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
98     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
99     velocity_threshold: float = 0.02
100     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
101     # enabled ? so the default path never imports torch / touches the GPU.
102     cue_flags: dict[str, bool] = Field(default_factory=dict)
103     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
104     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).

```

```

105     min_crossings: int = Field(default=0, ge=0)
106     # Served Gate B: when the person cue is on, drop windows with median in-court
players
107     # < this. 0 = OFF.
108     min_players: int = Field(default=0, ge=0)
109     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
110     court_polygon: Optional[list[list[float]]] = None
111     # Net line for the person two-sided-motion split (person features only ? the shuttle
112     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
113     net_axis: Literal["x", "y"] = "y"
114     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
115
116
117     class IndexingConfig(BaseModel):
118         default_segmenter: str = "yolo_hybrid"
119         use_gpu: bool = True
120         motion_threshold: float = 0.08
121         min_segment_duration: float = 3.0
122         dead_time_merge_gap: float = 2.0
123         chunk_duration: float = 90.0
124         chunk_overlap: float = 10.0
125         detector_model: str = "fasterrcnn_resnet50_fpn"
126         detector_score_threshold: float = 0.5
127         yolo_velocity_threshold: float = 0.15
128         yolo_frame_skip: int = 3
129         yolo_tracker: str = "bytetrack.yaml"
130         yolo_prefetch_workers: int = 2
131         min_action_window_duration: float = 2.0
132         # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
= a window
133         # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
134         # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't
swallow
135         # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
136         # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13
137         # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
138         # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
139         max_window_duration: float = 6.0
140         window_min_split_gap: float = 0.5 # min internal gap (s) that qualifies as a split
point
141         # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
142         # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
143         # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
144         # Only cuts (recall can't drop). OFF by default until measured/promoted.
145         window_hard_cap: bool = False
146         # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
147         # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
148         # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
149         # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
150         # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
151         # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap

```

```

152      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
153      window_inactivity_end: float = 1.5
154      inactivity_rally_thresh: float = 0.20
155      inactivity_min_density: float = 0.30
156      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
157      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
158      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
159      # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
160      # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
161      # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
162      # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
163      # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
164      window_blob_fallback: bool = False
165      # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
166      # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
167      # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
168      # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
169      window_blob_factor: float = 4.0
170      log_yolo_candidates: bool = True
171      store_yolo_candidates: bool = True
172      ai_handoff_padding: float = 2.0
173      ai_max_workers: int = 5
174      # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
175      # chunks of a high-bitrate (4K) source blow past the provider upload cap
176      # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
177      # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
178      # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
179      # Matches gemini_refine's --clip-scale-width default.
180      ai_chunk_scale_width: int = 1280
181      # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
182      # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
183      # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
184      # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
185      # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
186      skip_ai_handoff: bool = False
187      # Optional debug knob: trim every video to the first N seconds before processing.
188      debug_duration: Optional[float] = None
189
190      tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
191      wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
192      fusion: FusionConfig = Field(default_factory=FusionConfig)
193
194      @field_validator("default_segmenter")
195      @classmethod
196      def segmenter_must_be_registered(cls, v: str) -> str:
197          # Registry check happens at runtime in main/segmenter selection.
198          # Here we just allow any string for forward compatibility.
199          return v
200

```

```

201     @model_validator(mode="after")
202     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
203         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
204         # step makes `while t < duration: t += step` loop forever, growing memory before
any
205         # GPU work. Reject the bad config at load time instead.
206         if self.chunk_overlap >= self.chunk_duration:
207             raise ValueError(
208                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
209                 f"({self.chunk_duration}); otherwise chunking never advances."
210             )
211         return self
212
213
214     class OutputConfig(BaseModel):
215         default_highlights_name: str = "highlights.mp4"
216         db_name: str = "sports_indexer.db"
217         # Directory where the DB, highlight reels, and processed clips are written.
218         # The CLI's --output-dir overrides this at startup (see backend/main.py:main).
219         output_dir: str = "output"
220
221
222     class WatermarkConfig(BaseModel):
223         """Branding overlay burned into each highlight clip during the per-clip re-encode
224         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
225         (under `stitching.watermark` in config.json) to turn it off. The text is fully
226         configurable. The concat stage stays stream-copy (-c copy)."""
227         enabled: bool = True
228         text: str = "Generated by Khelsutra.guru"
229         logo_path: str = "" # optional transparent PNG overlay; "" = no logo
230         font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family
below
231         font: str = "Sans" # fontconfig family used when font_path is empty
232         font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of
frame height
233         opacity: float = Field(default=0.85, ge=0.0, le=1.0)
234         box: bool = True # faint backing box behind the text for legibility
235         margin: int = Field(default=24, ge=0) # px inset from the chosen corner
236         position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
237
238
239     class StitchingConfig(BaseModel):
240         begin_padding: float = 0.5
241         end_padding: float = 0.5
242         use_cache: bool = False
243         min_shot_count: int = 1
244         # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
245         # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
246         # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
247         # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
248         # local detector over-fires and its false positives are mostly SHORT (motion blips,
249         # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
250         # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
251         min_rally_duration: float = Field(default=0.0, ge=0.0)
252         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
253
254

```

```

255 class LoggingConfig(BaseModel):
256     """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
257     # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
258     # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
259     # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
260     # them to WARNING; set true to restore full chatter for request-flow debugging.
261     verbose_http: bool = False
262
263
264 class SecurityConfig(BaseModel):
265     # When set, /api/process video_path must resolve under this directory (hosted/tenant
266     # mode). Default None preserves the single-user local 'any local file' workflow.
267     allowed_video_root: Optional[str] = None
268
269     # --- Chunked upload (B2, PR-2) ---
270     # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
271     # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
272     # contain upload_dir or /api/process will reject the uploaded paths.
273     upload_dir: str = "output/uploads"
274     # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
275     # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
276     max_upload_mb: int = 4096
277     max_part_mb: int = 95
278     allowed_upload_extensions: List[str] = ["mp4", "mov"]
279
280
281 class AppConfig(BaseModel):
282     """Root configuration object.
283
284     All runtime configuration should be accessed via an instance of this class
285     (or the loader) rather than raw dict lookups.
286     """
287
288     sport: Sport = "badminton"
289     ai_provider: str = "gemini"
290     # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
flagship;
291     # the alias never deprecates underneath us. Eval reports record the run date
alongside.
292     ai_model: str = "gemini-flash-latest"
293
294     validation: ValidationConfig = Field(default_factory=ValidationConfig)
295     indexing: IndexingConfig = Field(default_factory=IndexingConfig)
296     output: OutputConfig = Field(default_factory=OutputConfig)
297     stitching: StitchingConfig = Field(default_factory=StitchingConfig)
298     security: SecurityConfig = Field(default_factory=SecurityConfig)
299     logging: LoggingConfig = Field(default_factory=LoggingConfig)
300
301     # Allow extra keys during transition so existing config.json files do not break.
302     model_config = {
303         "extra": "allow",
304         "validate_assignment": True,
305     }
306
307     # Temporary migration shim for legacy dict-style access (config.get("section",
{ }).get(...)).
308     # Will be removed once all call sites (validators, segmenters, etc.) move to typed
309     # attribute access / the typed config. A nested *section* is returned as a plain
dict
310     # via model_dump, so legacy chained access keeps working ? returning the Pydantic
311     # section model instead breaks it (a model has no .get), which bit every validator
on
312     # real runs. Scalar leaves are returned as-is.
313     def get(self, key: str, default: Any = None) -> Any:
314         _MISSING = object()

```

```

315         val: Any = _MISSING
316     if key in type(self).model_fields or hasattr(self, key):
317         val = getattr(self, key, _MISSING)
318     if val is _MISSING:
319         # Legacy convenience: find a leaf key inside any nested section. Sections
320         # discovered dynamically from the model fields, so a future top-level
321         # (e.g. `audio`, `jobs`) is covered automatically without editing a
322         # hardcoded list.
323         for fname in type(self).model_fields:
324             section = getattr(self, fname, None)
325             if isinstance(section, BaseModel) and hasattr(section, key):
326                 val = getattr(section, key)
327                 break
328     if val is _MISSING:
329         return default
330     if isinstance(val, BaseModel):
331         return val.model_dump(mode="json")
332     return val
333
334 def __getitem__(self, key: str) -> Any:
335     val = self.get(key)
336     if val is None and key not in type(self).model_fields:
337         # Re-raise like a real dict for unknown top-level
338         raise KeyError(key)
339     return val
340
341 # Convenience type alias for callers that want to annotate
342 Config = AppConfig
343

```

7. assistant (2026-06-19T15:52:14.118Z)

Now let me verify the remaining details. Let me check the §10 P2 line references and the rally_gate.py docstring to confirm the Gate B "future" claim context.

8. user (2026-06-19T15:52:15.970Z)

```

13-so the median sits near the net), then counts sign changes of (coord - net) across
14-the visible points inside a candidate window, with a deadband to ignore jitter.
15-
16:Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17-Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18-windows from trajectory_to_action_windows, or to fold into it later.
19-""

```

9. assistant (2026-06-19T15:52:27.499Z)

Now I have all the evidence I need. Let me verify each `stale\_text` is an exact, unique substring of the doc and assess each fix.

Let me confirm exact substrings appear verbatim in the doc by re-checking the specific lines.

Fix 1 (line 46): `machine, future`. The person cue in this plan is that named-but-unbuilt Gate B. appears verbatim at line 46. Code confirms: `fusion\_hybrid.py` implements Gate B (person path, `\_select\_for\_handoff` min_players at line 166, `\_person\_features` at 111-139). §10 P2 (line 271) marks FusionSegmenter DONE. So "named-but-unbuilt" is stale. However, the corrected_text has unbalanced markdown emphasis: `is that Gate B ? now built (...) see §10 P2. let me check it. Original: `is that named-but-unbuilt Gate B.`. The proposed correction keeps the `...` wrapping. It's accurate that it's built default-OFF + unmeasured. I'll confirm with a minor revision to keep voice tight.

****Fix 2**** (line 116): `?` (?2 in-court players ? recent track) (Gate B ? the person cue, NEW)`
? verbatim at line 116. Built, so "NEW" is stale. Confirm.

****Fix 3**** (line 178): the `shuttle\_player\_colocation` table row "NEW (needs the person cue)".
Built in `fusion\_features.py:168`, in `PERSON\_FEATURE\_NAMES`. Confirm.

****Fix 4**** (lines 141-143): person-features-built. Confirm but check the exact multi-line match
and emphasis.

****Fix 5**** (line 177): `net\_crossings` row "eval-only ? wire to prod". Code: `min\_crossings`
applied in `\_select\_for\_handoff` line 164. §1 (lines 61-62) says served Gate A wiring exists.
Confirm.

Let me verify the exact text of the multi-line stale_text for Fix 4 and the table rows for Fixes
3 and 5 are precise.

10. user (2026-06-19T15:52:28.307Z)

```
133     **Person features to add (~4, deliberately few):**
134
135     | Feature | Meaning | Rally signal |
136     |---|---|---|
137     | `players_in_court` | median count of persistent in-court tracks | ? 2 (singles) / 4
138     | `two_sided_motion` | are moving players on both net-halves? | rally = two-sided;
139     | `mean_player_motion` | aggregate normalized player velocity | rally = sustained;
140     | `in_court_ratio` | fraction of frames with ?2 in-court players | track stability vs
141     | `shuttle_player_colocation` | fraction of frames the shuttle sits inside/adjacent to a
142     | person box (hand region) | held = high; in-flight = mostly free space? the direct held-
143     | shuttle discriminator |
144     |
145     |
146     |
147     |
148     |
149     |
150     |
151     |
152     |
153     |
154     |
155     |
156     |
157     |
158     |
159     |
160     |
161     |
162     |
163     |
164     |
165     |
166     |
167     |
168     |
169     |
170     |
171     |
172     |
173     |
174     |
175     |
176     |
177     |
178     |
179     |
180     |
181     |
182     |
183     |
184     |
185     |
186     |
187     |
188     |
189     |
190     |
191     |
192     |
193     |
194     |
195     |
196     |
197     |
198     |
199     |
200     |
201     |
202     |
203     |
204     |
205     |
206     |
207     |
208     |
209     |
210     |
211     |
212     |
213     |
214     |
215     |
216     |
217     |
218     |
219     |
220     |
221     |
222     |
223     |
224     |
225     |
226     |
227     |
228     |
229     |
230     |
231     |
232     |
233     |
234     |
235     |
236     |
237     |
238     |
239     |
240     |
241     |
242     |
243     |
244     |
245     |
246     |
247     |
248     |
249     |
250     |
251     |
252     |
253     |
254     |
255     |
256     |
257     |
258     |
259     |
260     |
261     |
262     |
263     |
264     |
265     |
266     |
267     |
268     |
269     |
270     |
271     |
272     |
273     |
274     |
275     |
276     |
277     |
278     |
279     |
280     |
281     |
282     |
283     |
284     |
285     |
286     |
287     |
288     |
289     |
290     |
291     |
292     |
293     |
294     |
295     |
296     |
297     |
298     |
299     |
300     |
301     |
302     |
303     |
304     |
305     |
306     |
307     |
308     |
309     |
310     |
311     |
312     |
313     |
314     |
315     |
316     |
317     |
318     |
319     |
320     |
321     |
322     |
323     |
324     |
325     |
326     |
327     |
328     |
329     |
330     |
331     |
332     |
333     |
334     |
335     |
336     |
337     |
338     |
339     |
340     |
341     |
342     |
343     |
344     |
345     |
346     |
347     |
348     |
349     |
350     |
351     |
352     |
353     |
354     |
355     |
356     |
357     |
358     |
359     |
360     |
361     |
362     |
363     |
364     |
365     |
366     |
367     |
368     |
369     |
370     |
371     |
372     |
373     |
374     |
375     |
376     |
377     |
378     |
379     |
380     |
381     |
382     |
383     |
384     |
385     |
386     |
387     |
388     |
389     |
390     |
391     |
392     |
393     |
394     |
395     |
396     |
397     |
398     |
399     |
400     |
401     |
402     |
403     |
404     |
405     |
406     |
407     |
408     |
409     |
410     |
411     |
412     |
413     |
414     |
415     |
416     |
417     |
418     |
419     |
420     |
421     |
422     |
423     |
424     |
425     |
426     |
427     |
428     |
429     |
430     |
431     |
432     |
433     |
434     |
435     |
436     |
437     |
438     |
439     |
440     |
441     |
442     |
443     |
444     |
445     |
446     |
447     |
448     |
449     |
450     |
451     |
452     |
453     |
454     |
455     |
456     |
457     |
458     |
459     |
460     |
461     |
462     |
463     |
464     |
465     |
466     |
467     |
468     |
469     |
470     |
471     |
472     |
473     |
474     |
475     |
476     |
477     |
478     |
479     |
480     |
481     |
482     |
483     |
484     |
485     |
486     |
487     |
488     |
489     |
490     |
491     |
492     |
493     |
494     |
495     |
496     |
497     |
498     |
499     |
500     |
501     |
502     |
503     |
504     |
505     |
506     |
507     |
508     |
509     |
510     |
511     |
512     |
513     |
514     |
515     |
516     |
517     |
518     |
519     |
520     |
521     |
522     |
523     |
524     |
525     |
526     |
527     |
528     |
529     |
530     |
531     |
532     |
533     |
534     |
535     |
536     |
537     |
538     |
539     |
540     |
541     |
542     |
543     |
544     |
545     |
546     |
547     |
548     |
549     |
550     |
551     |
552     |
553     |
554     |
555     |
556     |
557     |
558     |
559     |
560     |
561     |
562     |
563     |
564     |
565     |
566     |
567     |
568     |
569     |
570     |
571     |
572     |
573     |
574     |
575     |
576     |
577     |
578     |
579     |
580     |
581     |
582     |
583     |
584     |
585     |
586     |
587     |
588     |
589     |
590     |
591     |
592     |
593     |
594     |
595     |
596     |
597     |
598     |
599     |
600     |
601     |
602     |
603     |
604     |
605     |
606     |
607     |
608     |
609     |
610     |
611     |
612     |
613     |
614     |
615     |
616     |
617     |
618     |
619     |
620     |
621     |
622     |
623     |
624     |
625     |
626     |
627     |
628     |
629     |
630     |
631     |
632     |
633     |
634     |
635     |
636     |
637     |
638     |
639     |
640     |
641     |
642     |
643     |
644     |
645     |
646     |
647     |
648     |
649     |
650     |
651     |
652     |
653     |
654     |
655     |
656     |
657     |
658     |
659     |
660     |
661     |
662     |
663     |
664     |
665     |
666     |
667     |
668     |
669     |
670     |
671     |
672     |
673     |
674     |
675     |
676     |
677     |
678     |
679     |
680     |
681     |
682     |
683     |
684     |
685     |
686     |
687     |
688     |
689     |
690     |
691     |
692     |
693     |
694     |
695     |
696     |
697     |
698     |
699     |
700     |
701     |
702     |
703     |
704     |
705     |
706     |
707     |
708     |
709     |
710     |
711     |
712     |
713     |
714     |
715     |
716     |
717     |
718     |
719     |
720     |
721     |
722     |
723     |
724     |
725     |
726     |
727     |
728     |
729     |
730     |
731     |
732     |
733     |
734     |
735     |
736     |
737     |
738     |
739     |
740     |
741     |
742     |
743     |
744     |
745     |
746     |
747     |
748     |
749     |
750     |
751     |
752     |
753     |
754     |
755     |
756     |
757     |
758     |
759     |
760     |
761     |
762     |
763     |
764     |
765     |
766     |
767     |
768     |
769     |
770     |
771     |
772     |
773     |
774     |
775     |
776     |
777     |
778     |
779     |
780     |
781     |
782     |
783     |
784     |
785     |
786     |
787     |
788     |
789     |
790     |
791     |
792     |
793     |
794     |
795     |
796     |
797     |
798     |
799     |
800     |
801     |
802     |
803     |
804     |
805     |
806     |
807     |
808     |
809     |
810     |
811     |
812     |
813     |
814     |
815     |
816     |
817     |
818     |
819     |
820     |
821     |
822     |
823     |
824     |
825     |
826     |
827     |
828     |
829     |
830     |
831     |
832     |
833     |
834     |
835     |
836     |
837     |
838     |
839     |
840     |
841     |
842     |
843     |
844     |
845     |
846     |
847     |
848     |
849     |
850     |
851     |
852     |
853     |
854     |
855     |
856     |
857     |
858     |
859     |
860     |
861     |
862     |
863     |
864     |
865     |
866     |
867     |
868     |
869     |
870     |
871     |
872     |
873     |
874     |
875     |
876     |
877     |
878     |
879     |
880     |
881     |
882     |
883     |
884     |
885     |
886     |
887     |
888     |
889     |
890     |
891     |
892     |
893     |
894     |
895     |
896     |
897     |
898     |
899     |
900     |
901     |
902     |
903     |
904     |
905     |
906     |
907     |
908     |
909     |
910     |
911     |
912     |
913     |
914     |
915     |
916     |
917     |
918     |
919     |
920     |
921     |
922     |
923     |
924     |
925     |
926     |
927     |
928     |
929     |
930     |
931     |
932     |
933     |
934     |
935     |
936     |
937     |
938     |
939     |
940     |
941     |
942     |
943     |
944     |
945     |
946     |
947     |
948     |
949     |
950     |
951     |
952     |
953     |
954     |
955     |
956     |
957     |
958     |
959     |
960     |
961     |
962     |
963     |
964     |
965     |
966     |
967     |
968     |
969     |
970     |
971     |
972     |
973     |
974     |
975     |
976     |
977     |
978     |
979     |
980     |
981     |
982     |
983     |
984     |
985     |
986     |
987     |
988     |
989     |
990     |
991     |
992     |
993     |
994     |
995     |
996     |
997     |
998     |
999     |
1000    |
```

11. assistant (2026-06-19T15:52:36.824Z)

All five `stale\_text` strings are confirmed verbatim and unique. The code evidence confirms
every claim:

- `FusionSegmenter` (fusion_hybrid.py) is built, registered `fusion\_hybrid`, default-OFF, with
Gate A (`\_select\_for\_handoff` line 164 via `min\_crossings`) and Gate B (line 166 via
`min\_players` + person cue).
- All 5 person features incl. `shuttle\_player\_colocation` are built in `fusion\_features.py`
(`PERSON\_FEATURE\_NAMES` line 194, `compute\_person\_features` line 200).
- `FusionConfig.min\_crossings`/`min\_players` exist in models.py (lines 105/108).
- §10 P2 (line 271) marks this DONE 2026-06-14; §1 (lines 61-62) already says served Gate A
exists. So §1's "named-but-unbuilt", the table rows' "NEW", and "eval-only ? wire to prod" all
contradict the same doc's later sections AND the code.

The corrections must honestly state "default-OFF + unmeasured on golden" (not "in production") ?
which they do, matching the code comments ("Lift to be measured", "mock-tested", line 281).

For Fix 1, I'll revise the corrected_text slightly: the proposed version has a markdown emphasis
imbalance (***is that Gate B ? now built (default-OFF + unmeasured on golden) as the` opens `***
then has an unclosed paren-bold). Let me tighten it to balanced, minimal text preserving voice.
All others are accurate as proposed.

12. user (2026-06-19T15:53:02.588Z)

Structured output provided successfully

13. assistant (2026-06-19T15:53:12.977Z)

All five proposed fixes survived verification. Summary of findings:

All five ``stale_text`` strings are exact, unique substrings of ``docs/MULTI_SIGNAL_FUSION_PLAN.md``, and every staleness claim is genuinely contradicted by current code:

- **Fix 1 (line 46, revise)** ? "named-but-unbuilt Gate B" is stale; Gate B is built in ``fusion_hybrid.py`` (``_select_for_handoff`` min_players gate line 166, ``_person_features`` lines 111-139). I revised the proposed ``corrected_text`` only to fix an unbalanced markdown-bold span and keep it minimal.
- **Fix 2 (line 116, confirm)** ? Gate B tagged "NEW"; built via ``FusionConfig.min_players`` (models.py:108) applied in ``_select_for_handoff`` (line 166).
- **Fix 3 (line 178, confirm)** ? ``shuttle_player_colocation`` tagged "NEW"; built in ``fusion_features.py:168``, in ``PERSON_FEATURE_NAMES`` (line 194), computed line 213.
- **Fix 4 (lines 141-143, confirm)** ? "Person features to add" framing stale; all five features built in ``fusion_features.py`` and persisted by ``fusion_hybrid.py::_person_features``.
- **Fix 5 (line 177, confirm)** ? ``net_crossings`` row says "eval-only ? wire to prod"; served wiring exists via ``FusionConfig.min_crossings`` (models.py:105) applied in ``_select_for_handoff`` (line 164).

Key honesty guardrail respected throughout: the corrections say "built, default-OFF, unmeasured on golden" ? never "in production" ? matching the code's own comments (``fusion_hybrid.py:281`` "Lift to be measured"; ``:116`` "mock-tested") and the doc's §10 P2 (DONE 2026-06-14, default-OFF) and §1 (lines 61-62) which already contradict these stale spots. Each row is an internal inconsistency where §1/the table lag §10 P2 and the shipped code.

I returned all five via StructuredOutput (one revise, four confirm).